

信号作业6

3-5(1)(3) 5-6(1) 7(1)(3) 8(1)(3)
9(1)(3) 10(1)(3) 11 13 14 15 16 17 18

3-5(1)

$$X(j\omega) = \int_{-\infty}^{+\infty} e^{-2j\omega t} e^{-j\omega t} dt$$

$$= \frac{e^{-2j\omega}}{j\omega + 2}$$

(3) $X(j\omega) = \int_{-\infty}^{+\infty} [\delta(t+\pi) + \delta(t-\pi)] e^{-j\omega t} dt$

$$= e^{-j\omega\pi} + e^{j\omega\pi}$$

$$= 2\cos\pi\omega$$

(1)

$$\frac{\sin\pi t}{\pi t} \xrightarrow{F} \text{rect}_{[-\pi, \pi]}(\omega)$$

$$\frac{\sin 2\pi t}{\pi t} \xrightarrow{F} \text{rect}_{[-2\pi, 2\pi]}(\omega)$$

$$\frac{\sin 2\pi(t-1)}{\pi(t-1)} \xrightarrow{F} \text{rect}_{[-2\pi, 2\pi]}(\omega) e^{-j\omega}$$

$$\frac{\sin\pi t}{\pi t} \cdot \frac{\sin 2\pi(t-1)}{\pi(t-1)} \xrightarrow{F} \frac{1}{2\pi} X_1(j\omega) * X_2(j\omega)$$

$$= \begin{cases} \frac{1+e^{-j\omega}}{2\pi j} & -3\pi < \omega < -\pi \\ -\frac{1+e^{j\omega}}{2\pi j} & \pi < \omega < 3\pi \\ 0 & \text{其它} \end{cases}$$

3-6(1)

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \pi [\delta(\omega+3\pi) + \delta(\omega-2\pi)] e^{j\omega t} d\omega$$

$$= \frac{1}{2} (e^{-j3\pi t} + e^{j2\pi t})$$

3-9(1)

$$x(t) \xrightarrow{F} X(j\omega)$$

$$x^*(t) \xrightarrow{F} X^*(-j\omega) = X(-j\omega) \neq X(j\omega)$$

复信号

$$x(-t) \xrightarrow{F} X(-j\omega)$$

$$x(t) \neq x(-t)$$

$$X(j\omega) \neq X(-j\omega)$$

非共轭偶信号

3-7(1)

$$x_1(t) = x(-t+3)$$

$$x(t) \xrightarrow{F} X(j\omega)$$

$$x(t+3) \xrightarrow{F} e^{3j\omega} X(j\omega)$$

$$x(-t+3) \xrightarrow{F} e^{-3j\omega} X(-j\omega)$$

(3) $x_2(t) = x(3-t)$

$$x(t) \xrightarrow{F} X(j\omega)$$

$$x_2(t) = x(3-t) \xrightarrow{F} \frac{1}{3} X(\frac{1}{3}j\omega)$$

(2)

$$x^*(t) \xrightarrow{F} X^*(-j\omega) = X(-j\omega) = -X(j\omega)$$

$$X(t) + X^*(t) = 0$$

$X(t)$ 是纯虚信号

$$x(-t) \xrightarrow{F} X(-j\omega) = -X(j\omega)$$

$$X(t) + X(-t) = 0$$

$X(t)$ 是奇信号

1-8(2)

$$e^{-at} u(t) \xrightarrow{F} \frac{1}{j\omega + a}$$

$$e^{-at} u(t) \cos\omega_0 t \xrightarrow{F} \frac{1}{2} \left[\frac{1}{j(\omega - \omega_0) + a} + \frac{1}{j(\omega + \omega_0) + a} \right]$$

(5)

$$\delta(t) \xrightarrow{F} 1$$

$$\delta'(t) \xrightarrow{F} j\omega \quad \delta(t+3) \xrightarrow{F} e^{3j\omega}$$

$$\delta(t+3) \xrightarrow{F} \frac{1}{2} e^{-3j\omega}$$

$$\delta'(t) + 2\delta(-2t+3) \xrightarrow{F} j\omega + e^{-j\frac{3}{2}\omega}$$

3-10(1)

$$\text{rect}_{[-3, 3]}(t) \xrightarrow{F} \text{rect}_{[-3, 3]}(\omega)$$

$$\frac{\sin 3\omega}{3\omega} = 2 \frac{\sin 3\omega}{\omega}$$

$$\text{rect}_{[-3, 3]}(t) e^{j\pi t} \xrightarrow{F} \frac{\sin 3(\omega - \pi)}{\omega - \pi}$$

$$x(t) = e^{j\pi t} [\mu(t+3) - \mu(t-3)]$$

(8)

$$\cos(\omega t + \frac{\pi}{2}) \xrightarrow{F} \frac{\pi}{2} e^{j\frac{\pi}{2}} [\delta(\frac{\omega}{2}-1) + \delta(\frac{\omega}{2}+1)]$$

$$\frac{\pi}{2} e^{j\frac{\pi}{2}} [\delta(\frac{\omega}{2}-1) + \delta(\frac{\omega}{2}+1)] \xrightarrow{F} 2\pi \cos(\omega t - \frac{\pi}{2})$$

$$\cos(\omega t - \frac{\pi}{2}) \xrightarrow{F} \frac{\pi}{2} e^{-j\frac{\pi}{2}} [\delta(\frac{\omega}{2}-1) + \delta(\frac{\omega}{2}+1)]$$

$$\frac{\pi}{2} e^{-j\frac{\pi}{2}} [\delta(\frac{\omega}{2}-1) + \delta(\frac{\omega}{2}+1)] \xrightarrow{F} \cos(2\omega t + \frac{\pi}{2}) \cdot 2\pi$$

$$X(t) = \frac{1}{4} \frac{\pi}{2} e^{-j\frac{\pi}{2}} [\delta(\frac{\omega}{2}-1) + \delta(\frac{\omega}{2}+1)]$$

$X(t)$ 为

$$\cos 2t \xrightarrow{F} \pi [\delta(\omega-2) + \delta(\omega+2)]$$

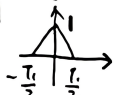
$$\cos(2t - \frac{\pi}{2}) \xrightarrow{F} \pi e^{-j\frac{\pi}{2}} [\delta(\omega-2) + \delta(\omega+2)]$$

对称性质

$$\pi e^{-j\frac{\pi}{2}} [\delta(\omega-2) + \delta(\omega+2)] \xrightarrow{F} \cos 2(t - \frac{\pi}{2})$$

$$X(t) = \frac{1}{2} e^{-j\frac{\pi}{2}} [\delta(t-2) + \delta(t+2)]$$

$$g(t) = \frac{1}{2} [e^{-j\frac{\pi}{2}} \delta(t-2) + e^{j\frac{\pi}{2}} \delta(t+2)]$$



$$X(t) = g(t) \cos(\omega_0 t)$$

$$X(j\omega) = \frac{\pi}{2\pi} [C_1 j(\omega - \omega_0) + C_1 j(\omega + \omega_0)]$$

$$= \frac{T_1}{4} \left\{ \text{Sa}[\frac{(\omega + \omega_0)T_1}{4}] + \text{Sa}[\frac{(\omega - \omega_0)T_1}{4}] \right\}$$

$$3-18 \quad y(t) \xrightarrow{F} E T \text{Sa}(\frac{T}{2}\omega)$$

$$x(t) = y(t - \frac{T}{2}) \xrightarrow{F} E T e^{-j\frac{T}{2}\omega} \text{Sa}(\frac{T}{2}\omega)$$

$$X(j\omega) = E T e^{-j\frac{T}{2}\omega} \text{Sa}(\frac{T}{2}\omega)$$

3-13

$$X(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

$$X(t) + X(t-1) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \cdot e^{-jk\omega_0} + \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0} \cdot e^{-jk\omega_0 t}$$

$$= \sum_{k=-\infty}^{+\infty} [(a_k + a_{-k}) e^{-jk\omega_0}] e^{jk\omega_0 t}$$

傅里叶级数系数为 $(a_k + a_{-k}) e^{-jk\omega_0}$

3-14

$x_1(t) \neq x_1^*(t)$ 非偶

$x_1(t) \neq x_1^*(t)$ 复信号

$x_2(j\omega) = x_2^*(j\omega)$ 偶信号

$x_2^*(j\omega) = x_2(j\omega)$ 实信号

$$x_3(t) = \sum_{k=-15}^{15} j \sin(\frac{k\pi}{2}) e^{jk\frac{\pi}{2}t} = -x_3^*(t) \text{ 奇信号}$$

$$x_3^*(t) = \sum_{k=-15}^{15} (-j) \sin(\frac{k\pi}{2}) e^{-jk\frac{\pi}{2}t}$$

$$= \sum_{k=-15}^{15} j \sin(\frac{k\pi}{2}) e^{jk\frac{\pi}{2}t} = x_3(t) \text{ 实信号}$$

106

实信号 $x_2(t)$ 和 $x_3(t)$

偶信号 $x_1(t)$

$$3-15 \quad x(t) = a_0 + \sum_{k=1}^{\infty} a_k \cos(k\pi t)$$

$$11) x(1) = a_0 + 2a_1 \cos \pi t$$

$$(4) \Rightarrow a_0 + 2a_1 = 1$$

$$\text{或 } a_0 = 1, a_1 = 0 \quad \text{或 } a_0 = 0, a_1 = \frac{1}{2} \quad \text{满足要求 } x(t) = 1 \text{ 或 } x(t) = \sqrt{2} \cos \pi t$$

$$3-16 \quad A e^{t} u(t) \xrightarrow{F} \frac{A}{j\omega + 1}$$

$$X(j\omega) = \frac{A}{j\omega + 1} \quad \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)| d\omega = 1 \Rightarrow A = \sqrt{2}$$

$$X(j\omega) = \sqrt{2} \left(\frac{1}{j\omega + 1} - \frac{1}{j\omega - 1} \right)$$

$$X(t) = 2\sqrt{2} (e^{-t} - e^{t}) u(t)$$

$$3-17 \quad x(t) = x_o(t) + x_e(t)$$

$$x_e(t) = e^{-t} u(t)$$

$$t \leq 0 \text{ 时 } x_o(t) = -x_e(t) = -e^{-t} u(-t)$$

$$x_o(t) = \begin{cases} e^{-t} & t > 0 \\ 0 & t = 0 \\ -e^{-t} & t < 0 \end{cases}$$

$$x(t) = 2e^{-t} u(t)$$

$$3-18 \quad y(t) \xrightarrow{F} E T \text{Sa}(\frac{T}{2}\omega)$$

$$x(t) = y(t - \frac{T}{2}) \xrightarrow{F} E T e^{-j\frac{T}{2}\omega} \text{Sa}(\frac{T}{2}\omega)$$

$$X(j\omega) = E T e^{-j\frac{T}{2}\omega} \text{Sa}(\frac{T}{2}\omega)$$

$$(1) \theta(\omega) = -\frac{T}{2}\omega$$

$$(2) X(0) = E T$$

$$(3) X(0) \cdot 2\pi = \int_{-\infty}^{+\infty} X(j\omega) d\omega = 2\pi E$$

(4)

$$y(t) \xrightarrow{F} Y(j\omega) = X(j\omega) \frac{2\sin \omega}{\omega}$$

$$\int_{-\infty}^{+\infty} Y(j\omega) e^{j\omega t} d\omega = 2\pi y(t)$$

$$g(t) \xrightarrow{F} 2 \frac{\sin \omega}{\omega} \quad g(t) = \frac{t}{u(t+1)} - u(t-1)$$

$$y(t) = x(t) * g(t)$$

$$y(t) = \int_{-\infty}^{+\infty} x(t') g(t-t') dt'$$

$$= \begin{cases} E(t-1) & 1 < T \leq 3 \\ 2E & T > 3 \\ 0 & T \leq 1 \end{cases}$$

$$\int_{-\infty}^{+\infty} Y(j\omega) e^{j\omega t} d\omega = \begin{cases} 2\pi E(t-1) & 1 < T \leq 3 \\ 4\pi E & T > 3 \\ 0 & T \leq 1 \end{cases}$$

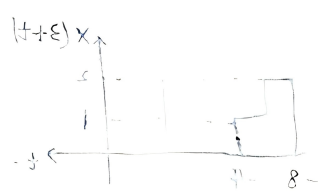
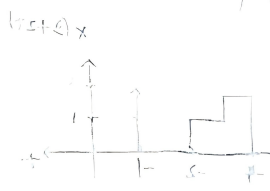
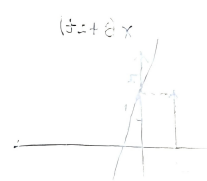
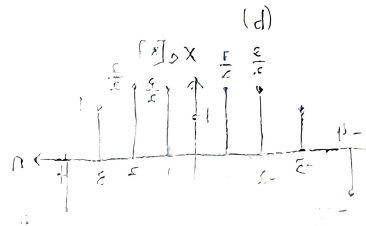
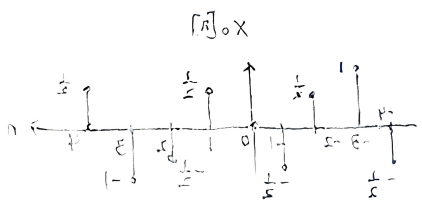
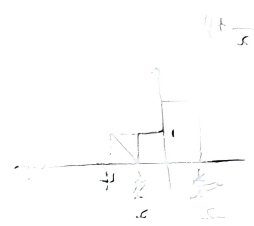
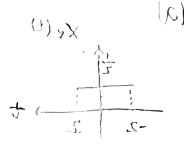
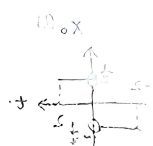
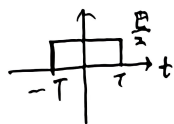
(5)

$$\int_{-\infty}^{+\infty} |X(\omega)|^2 d\omega = 2\pi \int_{-\infty}^{+\infty} |E|^2 dt$$

$$= 2\pi E^2 T$$

(6)

$$X_e(t) \xrightarrow{F} \text{Re}[X(\omega)]$$



$\frac{\pi}{\omega} = \frac{\pi \cdot 2\pi}{\omega} = T$
 时域脉宽 $\frac{\pi}{\omega} = \frac{\pi \cdot 2\pi}{\omega} = T$
 频域脉宽 $\frac{\pi}{\omega} = \frac{\pi \cdot 2\pi}{\omega} = T$

(8) 时域脉宽 $\frac{\pi}{\omega} = \frac{\pi \cdot 2\pi}{\omega} = T$ 频域脉宽 $\frac{\pi}{\omega} = \frac{\pi \cdot 2\pi}{\omega} = T$

$\omega = \frac{2\pi}{T} \quad \omega = \frac{2\pi}{T} \quad \omega = \frac{2\pi}{T}$

时域脉宽 $\frac{\pi}{\omega} = \frac{\pi \cdot 2\pi}{\omega} = T$ 频域脉宽 $\frac{\pi}{\omega} = \frac{\pi \cdot 2\pi}{\omega} = T$

时域	频域	时域	频域	时域
✓	✓	✗	✓	✗ (1)
✓	✓	✗	✓	✓ (2)
✗	✓	✓	✓	✗ (3)
✓	✓	✓	✓	✓ (4)
✓	✗	✓	✓	✓ (5)
✓	✗	✗	✗	✗ (6)